

Project Problem Identification

Problem Statement

Current surgical procedures often require highly invasive methods, leading to longer hospital stays, higher recovery demands on patients, and increased strain on healthcare resources. With the use of image-guided surgery tools such as the one we are enhancing, many procedures can be performed in a minimally invasive manner. This reduces the need for in-patient surgeries, shortens recovery time, and eases the workload on both doctors and hospitals while improving patient outcomes.

Initial Project Proposal

Literature / Market Review

Most image-guided surgery (IGS) tools on the market rely on external screens to display medical images during procedures. While effective, this approach requires surgeons to move their attention away from the patient to consult the display, potentially interrupting workflow and focus. Examples of existing systems include Medtronic's StealthStation platforms, Brainlab's Curve Image-Guided Surgery systems, and Navient's surgical navigation tools.

Our proposed solution differs in that it uses augmented reality (AR) to project surgical guidance directly into the surgeon's field of view, eliminating the need to look away from the operative site.

In addition, existing AR applications in surgery do not fully integrate real-time imaging modalities such as X-rays and ultrasound. Our approach aims to transform 2D scans into 3D volumes in real time and overlay them within the surgeon's AR headset, providing a more comprehensive and immediate understanding of the operative field.

AR technologies that have already been trialed in medical education and training have received highly positive feedback from users. These early successes highlight AR's potential to enhance surgical precision, efficiency, and overall workflow in clinical settings.

Product Vision Statement

For surgeons who require precise, minimally invasive guidance during complex procedures,
The Ctrl + Incision platform
Is an augmented reality surgical navigation system

That overlays real-time 3D reconstructions from 2D imaging sources (such as X-rays and ultrasound) directly into the surgeon's field of view.

Unlike conventional image-guided surgery systems that rely on external monitors and interrupt workflow,

Our product keeps the surgeon's focus on the patient while reducing invasiveness, recovery time, and strain on healthcare resources.

Target Audience / Stakeholders

The primary stakeholders are the surgeons and patients who will directly benefit from safer, less invasive procedures, and faster recovery. Secondary stakeholders include surgical staff, hospitals, and clinical technicians who will be responsible for integration and maintenance.

Success Criteria

For this project, our success criteria will be defined as:

- Less invasive approaches taken during surgeries when possible, resulting in faster procedures and recovery times
- Workload reduction on doctors and hospitals by turning in-patient surgeries into out-patient surgeries
- Less strain put on surgeons when using image-guided software tools by keeping their focus on the operative site
- Increased accuracy during surgeries

Risk Assessment

Technical Risks

- Accurate Registration & Tracking
 - Misalignment between AR overlays and the patient's anatomy could reduce surgical accuracy or even compromise patient safety.
- Latency & Real-Time Processing
 - Delays in generating 3D reconstructions from 2D scans (ultrasound, X-ray) could disrupt workflow or reduce surgeon trust.
- Hardware Limitations
 - Field of view, resolution, ergonomics, and comfort of AR glasses may hinder adoption.
- Integration with Existing Surgical Systems
 - Compatibility issues with hospital imaging devices and OR systems could delay adoption.
- Data Security & Privacy
 - Real-time patient imaging data may be vulnerable to breaches.

Organizational Risks

- Surgeon & Staff Adoption Resistance

- Surgeons may resist changing workflows or distrust AR overlays in high-stakes environments.
- Training Burden
 - Hospitals may view onboarding and training as too resource-intensive.
- Regulatory Approval & Compliance
 - Lengthy approval timelines (FDA, Health Canada, EMA) could delay time-to-market.
- Cost & Procurement Barriers
 - High upfront costs for hardware and integration may limit hospital adoption.
- Liability & Risk Management
 - Hospitals may fear liability if AR-guided surgery leads to complications.